



MISO DPP 2018 April Central Area Study Phase I Report

11/29/2019

**MISO
720 City Center
Drive Carmel
Indiana 46032**
<http://www.misoenergy.org>

Contents

1. Executive Summary	5
1.1. Project List.....	5
1.2. Total Network Upgrades	8
2. FERC Order 827 Compliance Review.....	12
3. Model Development and Study Assumptions.....	14
3.1. Base Case Models	14
3.2. Monitored Elements	14
3.3. Contingencies.....	15
3.4. Study Methodology	15
3.5. Performance Criteria	15
4. Thermal Analysis	15
5. Voltage Analysis	16
6. Stability Analysis	16
7. Short Circuit Analysis.....	16
8. Affected System Impact Study	16
9. Deliverability Analysis	16
9.1. Introduction.....	16
9.2. Determining the MW Restriction	16
9.3. Deliverability Study Results.....	16
9.3.1. J955.....	16
9.3.2. J956.....	17
9.3.3. J968.....	17
9.3.4. J974.....	17
9.3.5. J976.....	18
9.3.6. J979.....	18
9.3.7. J980.....	18
9.3.8. J985.....	18
9.3.9. J987.....	19
9.3.10. J991.....	19
9.3.11. J992.....	19
9.3.12. J993.....	19
9.3.13. J994.....	20
9.3.14. J995.....	20
9.3.15. J1016.....	20
9.3.16. J1021.....	21
9.3.17. J1022.....	21
9.3.18. J1025.....	21

9.3.19. J1026.....	21
9.3.20. J1027.....	22
9.3.21. J1028.....	22
9.3.22. J1031.....	22
9.3.23. J1033.....	23
9.3.24. J1034.....	23
9.3.25. J1039.....	24
9.3.26. J1055.....	24
9.3.27. J1058.....	24
9.3.28. J1059.....	25
9.3.29. J1063.....	25
9.3.30. J1067.....	25
9.3.31. J1069.....	25
9.3.32. J1074.....	26
9.3.33. J1087.....	26
9.3.34. J1094.....	27
9.3.35. J1096.....	27
9.3.36. J1102.....	27
9.3.37. J1107.....	28
9.3.38. J1111.....	28
9.3.39. J1112.....	29
9.3.40. J1115.....	29
9.3.41. J1123.....	29
9.3.42. J1139.....	29
9.3.43. J1145.....	30
9.3.44. J1152.....	30
9.3.45. J1161.....	30
9.3.46. J1180.....	31
9.3.47. J1189.....	31
10. Shared Network Upgrades Analysis	31
11. Cost Allocation	31
11.1. Cost Assumptions for Network Upgrades	31
11.2. Cost Allocation Methodology.....	32
Appendix A – Ameren System Impact Study Report (CEII)	33
Appendix B – Cost Allocation Summary (CEII)	33
Appendix C – MISO ERIS Analysis (CEII)	33
Appendix D – MISO Deliverability Analysis (CEII)	33
Appendix E – Network Upgrades Per Project (CEII)	33

Table 1: List of DPP April 2018 Central Area Phase I Projects	5
Table 2: Total Cost of Network Upgrades for DPP 2018 April Central Phase I Projects.....	8
Table 3: ERIS & NRIS Upgrades (Planning level cost estimates).....	9
Table 4: Shared Network Upgrades (Planning level cost estimates)	11
Table 5: FERC Order 827 Review Results	12
Table 6: NRIS Results for J956	17
Table 7: NRIS Results for J976	18
Table 8: NRIS Results for J985	18
Table 9: NRIS Results for J987	19
Table 10: NRIS Results for J994	20
Table 11: NRIS Results for J1016	20
Table 12: NRIS Results for J1026	21
Table 13: NRIS Results for J1027	22
Table 14: NRIS Results for J1028	22
Table 15: NRIS Results for J1033	23
Table 16: NRIS Results for J1034	23
Table 17: NRIS Results for J1039	24
Table 18: NRIS Results for J1059	25
Table 20: NRIS Results for J1074	26
Table 21: NRIS Results for J1087	26
Table 22: NRIS Results for J1094	27
Table 23: NRIS Results for J1107	28
Table 25: NRIS Results for J1123	29
Table 24: NRIS Results for J1145	30
Table 27: NRIS Results for J1161	30
Table 28: NRIS Results for J1189	31
Table 29: Maximum MW Impact and SNU Cost Allocations	31

1. Executive Summary

This report presents the results of a System Impact Study (SIS) performed to evaluate the interconnection of the generators in the DPP 2018 April Central Area Phase I (Central Area DPP I). The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren (AMIL, AMMO, ATXI), Duke Energy (DEI), Hoosier Energy (HE), Indianapolis Power & Light Company (IPL), Northern Indiana Public Service Company (NIPSCO), Southern Illinois Power Cooperative (SIPC), and Southern Indiana Gas & Electric Company (SIGE).

1.1. Project List

The original interconnection requests for DPP 2018 April Central Area had a total of 61 generation projects. Projects J957, J973, J983, J1012, J1017, J1018, J1019, J1035, J1066, J1080, J1116, J1118, and J1150 have withdrawn prior to the Phase I study. Therefore, there are 48 generation projects with a combined nameplate rating of 9348.23 MW (ERIS) & 9348.23 MW (NRIS). The detailed list of Central Area DPP I is shown below in Table 1, and the Phase I study was kicked off on April 22, 2019.

Table 1: List of DPP April 2018 Central Area Phase I Projects

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	POI
J955	Gas	ATXI	Sangamon County	IL	NRIS	1165 / 1040	Austin Substation 345kV bus
J956	Solar	AMMO	Ralls County	MO	NRIS	200	Spencer Creek 345kV Switching Station
J968	Wind	NIPSCO	Jasper County, White County	IN	NRIS	200	Reynolds 345kV Substation
J974	Wind	ATXI	Fulton County, Peoria County	IL	NRIS	250	Mapleridge 345kV Switching Station
J976	Solar	AMMO	Warren County	MO	NRIS	300	Montgomery – Enon Section of the Montgomery - Belleau 345kV Line
J979	Wind	ATXI	Christian County	IL	NRIS	200	Pana Substation 345kV Bus
J980	Wind	ATXI	Christian County	IL	NRIS	200	Pana Substation 345kV Bus
J985	Solar	AMMO	Scott County	MO	NRIS	120	Miner - Kelso 161kV Line
J987	Solar	AMMO	Montgomery County	MO	NRIS	100	Montgomery Substation 161kV Bus
J991	Solar	AMIL	Clay County	IL	NRIS	200	Xenia 345kV Switching Station
J992	Solar	DEI	Cass County	IN	NRIS	200	Walton 230kV Substation
J993	Solar	IPL	Boone County	IN	NRIS	200	Hortonville - Whitestown 345kV Line Tap
J994	Solar	AMMO	Callaway County	MO	NRIS	100	Guthrie Substation 161kV Bus

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	POI
J995	Solar	AMIL	Brown County, Jersey County	IL	NRIS	100	Jerseyville NW Substation 138kV Bus
J1016	Solar	SIGE	Gibson County	IN	NRIS	50	Francisco - Toyota Z75 138kV Line Tap
J1021	Wind	AMIL	Piatt County	IL	NRIS	200	Goose Creek 345kV Switchyard
J1022	Wind	AMIL	McLean County	IL	NRIS	150	Weedman Substation 138kV Bus
J1025	Wind	ATXI	Knox County	MO	NRIS	300	Zachary - Maywood 345kV Line
J1026	Wind	AMMO	Audrain County, Ralls County	MO	NRIS	400	Maywood - Spencer Creek 345kV Line
J1027	Solar	HE	Pike County	IN	NRIS	150	Ratts 161kV Substation
J1028	Solar	HE	Pike County	IN	NRIS	150	Ratts - Victory 161kV Line Tap
J1031	Wind	DEI	Henry County	IN	NRIS	150	Fall Creek - New Castle 138kV Line Tap
J1033	Battery Storage	AMMO	Stoddard County	MO	NRIS	50	Stoddard - Morely 161kV Line
J1034	Solar	AMMO	Stoddard County	MO	NRIS	225	Stoddard - Morely 161kV Line
J1039	Battery Storage	AMMO	Warren County	MO	NRIS	50	Enon - Montgomery 345kV Line
J1055	Wind	AMIL	Mason County	IL	NRIS	144	Mason Substation 138kV Bus
J1058	Solar	NIPSCO	Lake County	IN	NRIS	300	Schahfer-St. John 345kV Line Tap
J1059	Solar	AMMO	Cape Girardeau County	MO	NRIS	300	Kelso - Lutesville 345kV Line
J1063	Solar	DEI	Clinton County	IN	NRIS	300	New London - Frankfort 230kV Line Tap
J1067	Solar	NIPSCO	Jasper County, Pulaski County	IN	NRIS	240	Reynolds - Burr Oak 345kV Line
J1069	Wind	NIPSCO	Jasper County, Pulaski County	IN	NRIS	200	Reynolds 345kV Substation
J1074	Solar	SIGE	Gibson County	IN	NRIS	200	Francisco 138kV sub
J1087	Solar	AMMO	Scott County	MO	NRIS	200	Miner - Kelso 161kV Line
J1094	Solar	AMIL	Washington County	IL	NRIS	150	Prest 138kV Switching Station
J1096	Solar	AMIL	Saline County	IL	NRIS	150	Norris City North - Muddy 138kV Line
J1102	Solar	AMIL	Logan County	IL	NRIS	70	Fogarty 138kV Substation
J1107	Solar	AMMO	Cape Girardeau County	MO	NRIS	200	Kelso - Lutesville 345kV Line
J1111	Solar	SIPC	Jackson County	IL	NRIS	150	Campbell Hill - Jackson 161kV Line Tap

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	POI
J1112	Solar	NIPSCO	Kosciusko County	IN	NRIS	150	Leesburg 138kV Substation
J1115	Wind	AMIL	Macon County	IL	NRIS	200	Latham - Oreana 345kV Line
J1123	Wind	AMMO	Audrain County	MO	NRIS	150	Spencer Creek 345kV Switching Station
J1139	Solar	AMIL	Champaign County	IL	NRIS	150	Sidney Substation 138kV Bus
J1145	Solar	AMMO	Callaway County	MO	NRIS	250	Overton - (McCrede) - Montgomery 345kV Line
J1148	Gas	SIGE	Posey County	IN	NRIS	54	AB Brown 138kV Substation
J1152	Solar	IPL	Hancock County, Shelby County	IN	NRIS	200	Gwynneville - Sunnyside 345kV Line Tap
J1161	Solar	AMMO	Scott County	MO	NRIS	100.23	Miner - Kelso 161kV Line
J1180	Solar	AMIL	Clark County	IL	NRIS	75	Casey West - Sullivan 345kV Line
J1189	Battery Storage	DEI	Brown County, Martin County	IN	NRIS	5	Crane Solar 69kV Substation

1.2. Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2018 April Central Phase I reflects responsibilities for mitigating system impacts. The total cost of network upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies.

Table 2: Total Cost of Network Upgrades for DPP 2018 April Central Phase I Projects

Project	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)
	Thermal	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned				
a	b	c	d	e	f	g	h	i	j = b+c+d+f+g+i	\$4,000/MW	10% of (j) from Phase I-M2
J955	0	TBD	TBD	TBD	0	800,000	1,400,000	2,690,421	3,490,421	4,660,000	0.00
J956	584,132	TBD	TBD	TBD	6,813,119	700,000	800,000		8,097,251	800,000	9,725.10
J968	0	TBD	TBD	TBD	0	2,000,000			2,000,000	800,000	0.00
J974	0	TBD	TBD	TBD	0	800,000	1,000,000		800,000	1,000,000	0.00
J976	8,736	TBD	TBD	TBD	4,126,990	5,400,000	500,000		9,535,726	1,200,000	0.00
J979	0	TBD	TBD	TBD	0	800,000	1,400,000		800,000	800,000	0.00
J980	0	TBD	TBD	TBD	0	800,000	1,400,000		800,000	800,000	0.00
J985	4,483,520	TBD	TBD	TBD	632,414	10,800,000	800,000		15,915,934	480,000	1,111,593.40
J987	3,288	TBD	TBD	TBD	1,411,522	800,000	900,000		2,214,810	400,000	0.00
J991	0	TBD	TBD	TBD	0	800,000	1,000,000		800,000	800,000	0.00
J992	0	TBD	TBD	TBD	0	700,000			700,000	800,000	0.00
J993	0	TBD	TBD	TBD	0	10,425,000			10,425,000	800,000	242,500.00
J994	3,196	TBD	TBD	TBD	1,117,319	700,000	800,000		1,820,515	400,000	0.00
J995	2,068,029	TBD	TBD	TBD	0	700,000	800,000		2,768,029	400,000	0.00
J1016	0	TBD	TBD	TBD	5,407,864	10,859,226			16,267,090	200,000	1,426,709.00
J1021	0	TBD	TBD	TBD	0	800,000	1,000,000		800,000	800,000	0.00
J1022	0	TBD	TBD	TBD	0	700,000	800,000		700,000	600,000	0.00
J1025	30,682,330	TBD	TBD	TBD	0	10,800,000	1,000,000		41,482,330	1,200,000	2,948,233.00
J1026	2,580,738	TBD	TBD	TBD	14,655,541	10,800,000	1,000,000		28,036,279	1,600,000	1,203,627.90
J1027	0	TBD	TBD	TBD	8,976,955	2,100,000			11,076,955	600,000	507,695.50
J1028	0	TBD	TBD	TBD	9,198,791	7,900,000			17,098,791	600,000	1,109,879.10
J1031	0	TBD	TBD	TBD	0	10,000,000			10,000,000	600,000	400,000.00
J1033	393,803	TBD	TBD	TBD	267,309	4,000,000	400,000		4,661,112	200,000	266,111.20
J1034	1,772,197	TBD	TBD	TBD	1,204,520	4,000,000	400,000		6,976,717	900,000	0.00
J1039	2,912	TBD	TBD	TBD	687,832	5,400,000	500,000		6,090,744	200,000	409,074.40

J1055	827,721	TBD	TBD	TBD	0	1,156,000	881,000		1,983,721	576,000	0.00
J1058	0	TBD	TBD	TBD	0	15,000,000			15,000,000	1,200,000	300,000.00
J1059	685,772	TBD	TBD	TBD	1,628,303	11,100,000	1,000,000		13,414,075	1,200,000	141,407.50
J1063	30,215,000	TBD	TBD	TBD	18,750,000	13,000,000			61,965,000	1,200,000	4,996,500.00
J1067	0	TBD	TBD	TBD	0	15,000,000			15,000,000	960,000	540,000.00
J1069	0	TBD	TBD	TBD	0	2,000,000			2,000,000	800,000	0.00
J1074	0	TBD	TBD	TBD	21,631,455	4,162,154		260,744	26,054,353	800,000	1,805,435.30
J1087	7,557,326	TBD	TBD	TBD	1,054,567	10,800,000	800,000		19,411,893	800,000	1,141,189.30
J1094	0	TBD	TBD	TBD	978,775	5,300,000	800,000		6,278,775	600,000	27,877.50
J1096	0	TBD	TBD	TBD	0	8,000,000	800,000		8,000,000	600,000	200,000.00
J1102	0	TBD	TBD	TBD	0	700,000	800,000		700,000	280,000	0.00
J1107	457,237	TBD	TBD	TBD	1,086,350	11,100,000	1,000,000		12,643,587	800,000	464,358.70
J1111	8,926,000	TBD	TBD	TBD	823,116	4,500,000			14,249,116	600,000	824,911.60
J1112	0	TBD	TBD	TBD	0	4,000,000			4,000,000	600,000	0.00
J1115	0	TBD	TBD	TBD	0	10,800,000	1,000,000		10,800,000	800,000	280,000.00
J1123	876,197	TBD	TBD	TBD	5,109,839	700,000	800,000		6,686,036	600,000	68,603.60
J1139	0	TBD	TBD	TBD	0	700,000	800,000		700,000	600,000	0.00
J1145	8,471	TBD	TBD	TBD	2,872,757	11,000,000	1,000,000		13,881,228	1,000,000	388,122.80
J1152	0	TBD	TBD	TBD	0	10,425,000			10,425,000	800,000	242,500.00
J1161	3,895,894	TBD	TBD	TBD	529,728	8,420,150			12,845,772	400,920	883,657.20
J1180	0	TBD	TBD	TBD	0	11,800,000			11,800,000	300,000	880,000.00
J1189	0	TBD	TBD	TBD	284,936	0			284,936	20,000	8,493.60

Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. Energy Resource Interconnection Service (ERIS) Network Upgrades and Network Resource Interconnection Service (NRIS) Network Upgrades are shown in Table 3. Shared Network Upgrades are shown in Table 4.

Table 3: ERIS & NRIS Upgrades (Planning level cost estimates)

Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
Kelso – Miner 161kV Reconductor	AMMO	J1087, J1161, J985		5,250,000
Thorntown – Clarkshill – Potato Creek 69kV Rebuild	DEI	J1063		30,215,000
Thomas Hill – Adair 161kV Rebuild	AMMO	J1025		28,600,000
Cape – Heritage 161kV Rebuild	AMMO	J1161, J1087, J985, J1033, J1034, J1059, J1107		2,745,750

Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
Kelso 345/161kV Transformer Replacement	AMMO	J1161, J1087, J985, J1033, J1034		6,000,000
Miner – Sikeston 161kV Reconductor	AMMO	J1087, J1161, J985		5,250,000
Herleman 345/138kV Transformer Replacement	ATXI	J1026, J1123, J956, J1025		6,000,000
Marblehead N. 161/138kV Transformer Disconnect Switch and Bus Conductor	AMIL	J1025, J1026, J1039, J976, J1123, J1145, J956, J987, J994		150,000
Latham – Docket 138kV Rebuild	AMIL	J1055, J995		2,745,000
Campbell Hill – Bremen 69kV Rebuild	SIPC	J1111		2,655,000
Bremen – Evansville Tap 69kV Rebuild	SIPC	J1111		2,120,000
Evansville Tap – Sparta Tap 69kV Rebuild	SIPC	J1111		2,401,000
Campbell Hill 161/69kV Transformer Upgrade	SIPC	J1111		1,750,000
Taylorville South 138kV	AMIL	J995		150,000
J829 POI – Dresser 345kV Line	DEI		J1016, J1074, J1028, J1027, J1189	45,500,000
J1026 – Maywood 345kV Rebuild	AMMO		J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1111	45,000,000
New London – Kokomo 230kV Line	DEI		J1063	18,750,000

Table 4: Shared Network Upgrades (Planning level cost estimates)

Shared Network Upgrade	TO	Higher queued projects associated with SNU	Study projects associated with SNU	Cost of solution (\$)
Avena Tap – Ramsey 138kV		J811, J815, J848, J912	J955	7,500,000
Francisco – Duff 345kV		J800, J829, J842, J843	J1074	950,000

Note:

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.
- 3) Analysis performed shows there are two DPP-2018-APR Central projects for Shared Network Upgrade cost allocation.

2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, stated that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to all non-synchronous (wind, solar, and battery storage) projects included in the DPP 2018 April Central study cycle.

In this study, the power factor at the high-side of the generator substation for each inverter based project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that all projects meets the requirement to maintain 0.95 leading power factor, however, four projects do not meet the requirement to provide reactive power capability corresponding to 0.95 lagging power factor, as highlighted in red below in **Table 5**. Additional reactive support will be needed for these projects to meet the FERC requirement on reactive power capability prior to the completion of their GIA.

Table 5: FERC Order 827 Review Results

FERC Order 827				Steady State (At Generator Substation)							
Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	p.f. (pu)	P (MW)	Q (MVar)	p.f. (pu)		
J1016	50.4	±16.434	1 x 7 MVar Cap	49.8	15.3	0.9559	49.6	-27.7	-0.8731	No	1.1
J1021	200.0	±65.792	4 x 15 MVar Cap	195.9	95.8	0.8983	194.0	-121.5	-0.8475	Yes	
J1022	155.3	72.53 -66.14	2 x 10 MVar Cap	152.0	71.3	0.9053	150.3	-105.8	-0.8177	Yes	
J1025	319.0	±104.835	1 x 50 MVar Cap	306.0	134.9	0.9150	301.8	-149.9	-0.8956	Yes	
J1026	413.6	±135.924	1 x 65 MVar Cap	390.2	143.9	0.9382	379.0	-247.4	-0.8374	Yes	
J1027	150.0	±25.144	N/A	148.1	-4.6	0.9995	147.7	-61.3	-0.9236	No	53.3
J1028	150.0	±25.144	N/A	147.8	-4.8	0.9995	147.4	-62.2	-0.9213	No	53.4
J1031	150.0	±49.275	N/A	148.3	26.2	0.9848	147.5	-88.3	-0.8580	No	22.5
J0133 & J1034	275.0	±84.5	1 x 4 MVar Cap	270.9	37.2	0.9907	268.8	-163.4	-0.8545	No	51.8
J1039	50.0	±30	1 x 4 MVar Cap	49.6	24.8	0.8944	49.3	-45.0	-0.7386	Yes	
J1055	144.0	67.68 -80	1 x 7.5 MVar Cap	132.8	55.6	0.9224	125.6	-128.5	-0.6990	Yes	

J1058	300.0	±146.4	1 x 4 MVar Cap	293.5	90.4	0.9557	289.6	-244.0	-0.7647	No	6.1
J1059	300.0	±146.4	1 x 4 MVar Cap	293.6	93.3	0.9530	289.5	-242.5	-0.7666	No	3.2
J1063	302.4	±146.46	2 x 4 MVar Cap	296.2	99.4	0.9480	291.2	-248.6	-0.7605	Yes	
J1067	240.0	±97.9	2 x 17 MVar Cap	236.9	91.6	0.9327	235.4	-161.4	-0.8248	Yes	
J1069	200.0	±81.6	2 x 14 MVar Cap	196.6	80.8	0.9249	195.1	-130.3	-0.8316	Yes	
J1074	200.0	±36.925	N/A	197.2	-1.8	1.0000	196.6	-84.8	-0.9182	No	66.6
J1087	200.0	±36.925	N/A	197.2	1.6	1.0000	196.6	-84.7	-0.9184	No	63.2
J1094	150.0	±25.144	N/A	148.0	-3.0	0.9998	147.6	-58.5	-0.9296	No	51.6
J1096	150.0	±25.144	N/A	148.0	-2.9	0.9998	147.5	-59.7	-0.9270	No	51.5
J1102	70.0	±28.555	1 x 8 MVar Cap	68.7	27.5	0.9284	68.2	-43.9	-0.8409	Yes	
J1107	200.0	±87.7	2 x 14 MVar Cap	197.4	82.2	0.9232	196.2	-142.2	-0.8097	Yes	
J1111	150.0	±25.144	N/A	148.0	-2.4	0.9999	147.7	-57.6	-0.9317	No	51.0
J1112	153.3	±74.2	N/A	151.5	57.4	0.9351	150.8	-101.3	-0.8301	Yes	
J1115	200.0	±65.793	4 x 15 MVar Cap	195.8	112.5	0.8671	194.3	-106.3	-0.8773	Yes	
J1123	151.8	±49.888	3 x 15 MVar Cap	149.0	81.8	0.8766	148.1	-82.9	-0.8726	Yes	
J1139	151.2	±95.3400	N/A	149.4	74.0	0.8961	148.6	-127.8	-0.7582	Yes	
J1145	250.0	±119.07	N/A	246.5	68.0	0.9640	244.4	-200.6	-0.7730	No	13.0
J1152	200.0	±94.3	N/A	196.6	52.1	0.9666	194.4	-163.8	-0.7647	No	12.5
J1161	100.2	±48.551	N/A	98.8	35.6	0.9408	98.2	-68.2	-0.8213	Yes	
J1180	75.0	±65.233	4 x 6 MVar Cap 1 x 6 MVar Inductor	73.8	86.8	0.6478	73.2	-86.9	-0.6442	Yes	
J1189	5.00	0.0	N/A	4.9	0.5	0.9948	4.9	0.5	-0.9948	No	1.1
J956	200.6	±97	N/A	197.9	68.1	0.9456	196.9	-141.3	-0.8124	Yes	
J968	200.0	66 -64	2 x 12 MVar Cap	196.7	62.6	0.9529	195.6	-105.1	-0.8809	No	2.1

J974	250.0	82.5 -80	2 x 9 MVar Cap	244.0	61.7	0.9695	241.9	-137.2	-0.8698	No	18.5
J976	300.0	±146.4	2 x 4 MVar Cap	293.7	98.3	0.9483	289.9	-238.9	-0.7717	Yes	
J979	200.0	66 -64	2 x 22 MVar Cap	196.5	80.9	0.9247	195.1	-118.0	-0.8557	Yes	
J980	200.0	66 -64	2 x 22 MVar Cap	196.5	80.7	0.9250	195.0	-119.1	-0.8534	Yes	
J985	120.0	±48.95	6 x 10.2 MVar Cap	118.0	99.1	0.7658	117.3	-76.4	-0.8379	Yes	
J987	100.0	±44.24	2 x 7 MVar Cap	98.8	42.2	0.9196	98.2	-70.1	-0.8139	Yes	
J991	200.0	±88.48	2 x 14 MVar Cap	197.5	85.6	0.9175	196.4	-139.9	-0.8145	Yes	
J992	200.0	±88.48	2 x 14 MVar Cap	197.5	82.7	0.9224	196.2	-146.3	-0.8017	Yes	
J993	200.0	±88.48	2 x 14 MVar Cap	197.3	81.1	0.9249	196.0	-145.3	-0.8033	Yes	
J994	100.0	±44.24	2 x 7 MVar Cap	98.8	40.8	0.9243	98.0	-73.5	-0.8000	Yes	
J995	100.0	±44.24	2 x 7 MVar Cap	98.8	43.1	0.9166	98.0	-72.1	-0.8055	Yes	

3. Model Development and Study Assumptions

3.1. Base Case Models

The origin of the DPP 2018 April Central models is the MTEP 18 models with the Bench Cases including all pre-queued projects and associated network upgrades, while the Study Cases contain all of the interconnection requests in DPP 2018 April Central Phase I, in addition to all the facilities in the Bench Cases.

- Bench Cases
 - APR18-2023Shoulder-Bench_updates_8-2-19.raw
 - APR18-2023SUM-Bench_Updated8-2-19.raw
- Study Cases
 - APR18-2023Shoulder-Study_Updated_8-2-19.raw
 - APR18-2023Shoulder-Study_Charging_8-2-19.raw
 - APR18-2023SUM-Study_Charging_8-2-19.raw
 - APR18-2023SUM-Study_Updated_8-2-19.raw

3.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

3.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2018 April Central generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 33.12.1 and TARA version 1802b.

3.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as Injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner FERC filed Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the Deliverability algorithm, to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

4. Thermal Analysis

The thermal analysis results for 2018 April show generator projects J955, J956, J976, J985, J987, J994, J995, J1025, J1026, J1033, J1034, J1039, J1055, J1059, J1063, J1087, J1102, J1111, J1112, J1123, J1145, and J1161 causing constraints. The details pertaining to the thermal analysis can be found in Appendix A – Ameren System Impact Study Report and Appendix D – MISO ERIS Analysis.

5. Voltage Analysis

The voltage analysis results for 2018 April show that the study generators do not cause any voltage violations. The details pertaining to the voltage analysis can be found in MISO ERIS Analysis (CEII).

6. Stability Analysis

Stability analysis will be performed in Phase II of the 2018 APRIL Central cycle.

7. Short Circuit Analysis

Short circuit analysis will be performed in Phase II of the 2018 April Central cycle.

8. Affected System Impact Study

Affected System Impact Study will be performed in Phase II of the 2018 April Central cycle.

9. Deliverability Analysis

9.1. Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource (ER) project or to proceed with the system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

9.2. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator's output is not deliverable.

9.3. Deliverability Study Results

The limiting constraints (mon-con pairs) seen in the deliverability analysis for the 2018 Summer case are summarized in Appendix A – Ameren System Impact Study Report Deliverability Analysis (CEII).

9.3.1. J955

This generator is found to be fully deliverable for 1165 MW without any network upgrades.

9.3.2. J956

This generator is determined to be fully deliverable for 189.26 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 6: NRIS Results for J956

J956 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERS and case assumptions)		189.26 MW (94.63%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	200	41.81%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	6,813,119	45,000,000

9.3.3. J968

This generator is found to be fully deliverable for 200 MW without any network upgrades.

9.3.4. J974

This generator is found to be fully deliverable for 250 MW without any network upgrades.

9.3.5. J976

This generator is determined to be fully deliverable for 283.88 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 7: NRIS Results for J976

J976 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIIS and case assumptions)		283.88 MW (94.63%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distrib ution Factor	Constraint in ERIIS Analysis?	Projects Associate d with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	300	16.89 %	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	4,126,990	45,000,000

9.3.6. J979

This generator is found to be fully deliverable for 200 MW without any network upgrades.

9.3.7. J980

This generator is found to be fully deliverable for 200 MW without any network upgrades.

9.3.8. J985

This generator is determined to be fully deliverable for 113.55 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 8: NRIS Results for J985

J985 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIIS and case assumptions)		113.55 MW (94.63%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	120	6.47%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	632,414	45,000,000

9.3.9. J987

This generator is determined to be fully deliverable for 94.63 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 9: NRIS Results for J987

J987 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERS and case assumptions)		94.63 MW (94.63%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	100	17.32%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	1,411,522	45,000,000

9.3.10. J991

This generator is found to be fully deliverable for 200 MW without any network upgrades.

9.3.11. J992

This generator is found to be fully deliverable for 200 MW without any network upgrades.

9.3.12. J993

This generator is found to be fully deliverable for 200 MW without any network upgrades.

9.3.13. J994

This generator is determined to be fully deliverable for 94.63 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 10: NRIS Results for J994

J994 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIIS and case assumptions)		94.63 MW (94.63%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	100	13.71%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	1,117,319	45,000,000

9.3.14. J995

This generator is found to be fully deliverable for 100 MW without any network upgrades.

9.3.15. J1016

This generator is determined to be fully deliverable for 0 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 11: NRIS Results for J1016

J1016 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIIS and case assumptions)		0 MW (0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J829 – Dresser 345kV Line	50	10.97%	No	-	J1016, J1074, J1028, J1027, J1189	5,407,864	45,500,000

9.3.16. J1021

This generator is found to be fully deliverable for 200 MW without any network upgrades.

9.3.17. J1022

This generator is found to be fully deliverable for 150 MW without any network upgrades.

9.3.18. J1025

This generator is found to be fully deliverable for 300 MW without any network upgrades.

9.3.19. J1026

This generator is determined to be fully deliverable for 378.51 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 12: NRIS Results for J1026

J1026 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIS and case assumptions)				378.51 MW (94.63%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	400	44.96%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	14,655,541	45,000,000

9.3.20. J1027

This generator is determined to be fully deliverable for 0 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 13: NRIS Results for J1027

J1027 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIS and case assumptions)				0 MW (0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J829 – Dresser 345kV Line	150	6.07%	No	-	J1016, J1074, J1028, J1027, J1189	8,976,955	45,500,000

9.3.21. J1028

This generator is determined to be fully deliverable for 0 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 14: NRIS Results for J1028

J1028 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIS and case assumptions)				0 MW (0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J829 – Dresser 345kV Line	150	6.22%	No	-	J1016, J1074, J1028, J1027, J1189	9,198,791	45,500,000

9.3.22. J1031

This generator is found to be fully deliverable for 150 MW without any network upgrades.

9.3.23. J1033

This generator is determined to be fully deliverable for 47.31 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 15: NRIS Results for J1033

J1033 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIIS and case assumptions)				47.31 MW (94.63%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	50	6.57%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	267,309	45,000,000

9.3.24. J1034

This generator is determined to be fully deliverable for 212.91 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 116: NRIS Results for J1034

J1034 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIIS and case assumptions)				212.91 MW (94.63%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	225	6.57%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	1,204,520	45,000,000

9.3.25. J1039

This generator is determined to be fully deliverable for 47.31 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 17: NRIS Results for J1039

J1039 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERS and case assumptions)		47.31 MW (94.63%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	50	16.88%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	687,832	45,000,000

9.3.26. J1055

This generator is found to be fully deliverable for 144 MW without any network upgrades.

9.3.27. J1058

This generator is found to be fully deliverable for 300 MW without any network upgrades.

9.3.28. J1059

This generator is determined to be fully deliverable for 283.88 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 18: NRIS Results for J1059

J1059 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIIS and case assumptions)		283.88 MW (94.63%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	300	6.66%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	1,628,303	45,000,000

9.3.29. J1063

This generator is determined to be fully deliverable for 294.14 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 19: NRIS Results for J1063

J1063 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIIS and case assumptions)		294.14 MW (98.05%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
New London – Kokomo 230kV Line	300	43.80%	No	-	J1063	18,750,000	18,750,000

9.3.30. J1067

This generator is found to be fully deliverable for 240 MW without any network upgrades.

9.3.31. J1069

This generator is found to be fully deliverable for 200 MW without any network upgrades.

9.3.32. J1074

This generator is determined to be fully deliverable for 0 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 19: NRIS Results for J1074

J1074 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIS and case assumptions)				0 MW (0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J829 – Dresser 345kV Line	200	10.97%	No	-	J1016, J1074, J1028, J1027, J1189	21,631,455	45,500,000

9.3.33. J1087

This generator is determined to be fully deliverable for 189.26 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 20: NRIS Results for J1087

J1087 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIS and case assumptions)				189.26 MW (94.63%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	200	6.47%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	1,054,567	45,000,000

9.3.34. J1094

This generator is determined to be fully deliverable for 141.94 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 21: NRIS Results for J1094

J1094 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERS and case assumptions)		141.94 MW (94.63%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	150	8.00%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	978,775	45,000,000

9.3.35. J1096

This generator is found to be fully deliverable for 150 MW without any network upgrades.

9.3.36. J1102

This generator is found to be fully deliverable for 70 MW without any network upgrades.

9.3.37. J1107

This generator is determined to be fully deliverable for 189.26 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 22: NRIS Results for J1107

J1107 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERS and case assumptions)		189.26 MW (94.63%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	200	6.67%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	1,086,350	45,000,000

9.3.38. J1111

This generator is determined to be fully deliverable for 141.94 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 24: NRIS Results for J1111

J1111 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERS and case assumptions)		141.94 MW (94.63%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associa ted with ERS Constra int	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	150	6.73%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	823,116	45,000,000

9.3.39. J1112

This generator is found to be fully deliverable for 150 MW without any network upgrades.

9.3.40. J1115

This generator is found to be fully deliverable for 200 MW without any network upgrades.

9.3.41. J1123

This generator is determined to be fully deliverable for 141.94 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 23: NRIS Results for J1123

J1123 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERS and case assumptions)		141.94 MW (94.63%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	150	41.80%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	5,109,839	45,000,000

9.3.42. J1139

This generator is found to be fully deliverable for 150 MW without any network upgrades.

9.3.43. J1145

This generator is determined to be fully deliverable for 236.57 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 24: NRIS Results for J1145

J1145 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIS and case assumptions)		236.57 MW (94.63%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	250	14.10%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	2,872,575	45,000,000

9.3.44. J1152

This generator is found to be fully deliverable for 200 MW without any network upgrades.

9.3.45. J1161

This generator is determined to be fully deliverable for 94.85 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 25: NRIS Results for J1161

J1161 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERIS and case assumptions)		94.85 MW (94.63%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J1026 – Maywood 345kV Rebuild	100.23	6.49%	No	-	J1145, J994, J987, J976, J1039, J956, J1123, J1026, J1033, J1034, J985, J1087, J1161, J1107, J1059, J1094, J1111	529,728	45,000,000

9.3.46. J1180

This generator is found to be fully deliverable for 75 MW without any network upgrades.

9.3.47. J1189

This generator is determined to be fully deliverable for 0 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. **Error! Reference source not found.** shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 26: NRIS Results for J1189

J1189 Deliverable (NRIS) Amount in 2018 Case: (Conditional on ERS and case assumptions)				0 MW (0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated with ERS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)
J829 – Dresser 345kV Line	5	5.78%	No	-	J1016, J1074, J1028, J1027, J1189	284,936	45,500,000

10. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) Analysis tests for Network Upgrades driven by higher queued interconnection projects was performed for this System Impact Study.

The maximum MW impacts and Shared Network Upgrade (SNU) cost allocations appear in Table 27.

Table 27: Maximum MW Impact and SNU Cost Allocations

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility (\$)
Avena – Ramsey Tap 138kV	DPP-2017-AUG	J811	16.7127	\$7,500,000	\$2,195,441
	DPP-2017-AUG	J815	4.16		\$546,473
	DPP-2017-AUG	J848	12.9955		\$1,707,137
	DPP-2017-AUG	J912	2.7445		\$360,528
	DPP-2018-APR	J955	10.24035		\$2,690,421
Francisco – Duff 345kV	DPP-2017-AUG	J800	17.65	\$950,000	\$116,355
	DPP-2017-AUG	J829	19.075		\$125,749
	DPP-2017-AUG	J842	40.18		\$264,880
	DPP-2017-AUG	J843	44.28		\$291,908
	DPP-2018-APR	J1074	22.922		\$151,109

11. Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

11.1. Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade identified in System Impact Study was provided by the corresponding transmission owning company.

11.2. Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

Cost Allocation Methodology for Thermal Constraints

1. With all Study Group generation projects dispatched in the Post Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any network upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$\text{Cost of NU} = \frac{\text{Project A cost portion of NU} \times \text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contrution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU is required.



Appendix A – Ameren System Impact Study Report (CEII)

Appendix B – Cost Allocation Summary (CEII)

Appendix C – MISO ERIS Analysis (CEII)

Appendix D – MISO Deliverability Analysis (CEII)

Appendix E – Network Upgrades Per Project (CEII)